

Comparison of Multilabel Classification Techniques for Emotion Recognition

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Abstract—Multi-label classification (MLC) has emerged as a critical area of supervised learning where each instance may be associated with multiple labels simultaneously. This paper presents a comprehensive comparative analysis of four transformation-based multi-label classification methods: Binary Relevance (BR), Classifier Chains (CC), Label Powerset (LP), and Random k-Labelsets (RAkEL). These methods are evaluated using four widely adopted classifiers: Naive Bayes, Logistic Regression, Random Forest, and Support Vector Machine (SVM) with linear kernel. Experiments are performed on the Emotion dataset, a recognized benchmark in text-based MLC tasks, using standard evaluation metrics including Hamming Loss, Exact Match Ratio, Micro-F1, and Macro-F1 scores. The results highlight trade-offs between predictive accuracy, scalability, and computational efficiency, providing insights for designing more effective MLC pipelines. The findings indicate that Random Forest with RAkEL achieves the highest performance with 0.4202 subset accuracy and 0.6737 micro-F1 score, while SVM-based methods show consistent performance across different transformation techniques.

Index Terms—Multi-label classification, Binary Relevance, Classifier Chains, Label Powerset, RAkEL, Emotion dataset, text classification, machine learning

I. INTRODUCTION

Multi-label classification (MLC) has emerged as a significant paradigm in supervised learning, where each instance may be associated with multiple labels simultaneously. Unlike traditional single-label or multi-class classification, where each instance corresponds to exactly one class, MLC captures the more complex and overlapping relationships often encountered in real-world scenarios. For example, in document classification, a single article may belong to multiple categories such as politics, economics, and health. Similarly, in music tagging, a song can simultaneously be labeled with genres like jazz, instrumental, and blues. These scenarios highlight the broader applicability of MLC compared to its single-label counterpart.

The need for robust MLC models has increased with the growing prevalence of data sources that inherently involve multiple co-occurring attributes. Emotion recognition, in particular, is one such domain where the identification of more than one emotional state is essential. For instance, a person may simultaneously experience emotions such as joy and surprise, or sadness and fear, which cannot be sufficiently modeled using single-label classification techniques. Effective

emotion recognition has a wide range of applications, including human-computer interaction, sentiment analysis, mental health monitoring, and personalized recommendation systems. These applications highlight the societal and technological importance of accurately modeling emotions in multi-label settings.

Several transformation-based approaches have been developed to address the challenges of MLC by reducing it into simpler learning tasks. Among these, Binary Relevance (BR) treats each label independently, Classifier Chains (CC) models sequential label dependencies, Label Powerset (LP) transforms the problem into a multi-class task by representing unique label combinations as single classes, and Random k-Labelsets (RAkEL) builds ensembles on random subsets of labels to capture dependencies more efficiently. Each of these methods has its strengths and limitations, often depending on the underlying classifier and dataset characteristics.

In this paper, we conduct a comprehensive comparative study of these four prominent transformation-based approaches combined with multiple classifiers, namely Naive Bayes, Logistic Regression, Random Forest, and Support Vector Machine (SVM) with a linear kernel. The experiments are performed on the widely recognized Emotion Dataset, a benchmark specifically designed for text-based multi-label classification. Unlike generic datasets, the Emotion dataset captures nuanced emotional dependencies that make it highly relevant for benchmarking MLC methods. The evaluation uses standard metrics such as Hamming Loss, Exact Match Ratio, and F1-Score to provide insights into trade-offs between predictive accuracy, computational efficiency, and the ability to model label dependencies.

The contributions of this study are threefold: (1) a systematic exploration of transformation-based MLC methods in the context of emotion recognition, (2) an empirical comparison of widely used classifiers under uniform experimental settings, and (3) practical insights into how different methods balance scalability, interpretability, and predictive performance. This analysis contributes to the broader effort of advancing effective multi-label learning pipelines for real-world applications.

II. LITERATURE REVIEW

Research on multi-label classification has grown rapidly, with a wide range of methods proposed to improve predictive accuracy and computational efficiency. Some provided an early survey categorizing methods into problem transformation and algorithm adaptation strategies. Problem transformation remains widely adopted due to its compatibility with general-purpose classifiers.

Binary Relevance (BR) is one of the simplest approaches, decomposing MLC into independent binary tasks. However, it neglects label dependencies, which can reduce effectiveness in correlated domains. To address this, Classifier Chains (CC) were introduced, where predictions of prior labels are used as features for subsequent classifiers, capturing dependencies but suffering from error propagation.

Label Powerset (LP), represents each unique label combination as a single class, enabling modeling of label interactions. Nevertheless, scalability issues arise when the label space is large. RAKEL mitigates this by training ensembles on random subsets of labels, striking a balance between dependency modeling and efficiency.

Classifier choice further impacts performance. Linear models like Logistic Regression excel in high-dimensional sparse text data, while Random Forest offers robustness in mixed-feature domains. Naive Bayes remains a strong baseline in text-based problems due to its probabilistic assumptions. SVMs, particularly linear variants, are widely applied in MLC because of their strong generalization capabilities.

Recent advances in automated emotion and multilabel understanding emphasize dataset quality, multimodal fusion, and ensemble architectures. Yalçın and Alisawi [5] address facial emotion recognition by introducing a carefully preprocessed and expanded FER2024 CK+ dataset and demonstrate that attention-augmented CNNs and transfer-based Efficient-NetB7-CNN substantially improve class-wise recall and overall accuracy on challenging FER benchmarks. Bilotti et al. [4] systematically compare single- and multi-input CNN fusion strategies on video-derived modalities and audio spectrograms, showing large accuracy gains. Nimmi et al. [3] demonstrate that ensembles of fine-tuned transformer encoders outperform individual transformers in crisis-related text classification tasks. Gross et al. [1] provide methodological perspectives on when direct MLC can outperform detection-based pipelines. Saleh [2] investigates imbalanced, multi-label social-media classification and reports ensemble benefits for skewed multi-label text tasks.

III. RESEARCH GAPS

Although multi-label classification (MLC) has received considerable attention in the last decade, several research gaps remain unaddressed that justify the need for further exploration. A review of existing studies highlights the following limitations:

- **Limited focus on emotion recognition:** While MLC techniques have been widely applied to domains such

as document categorization, image annotation, and bioinformatics, relatively fewer studies have emphasized the domain of emotion recognition. Emotional data poses unique challenges due to overlapping and co-occurring emotions, ambiguity in text semantics, and contextual dependencies.

- **Insufficient evaluation of transformation-based methods:** Most prior works evaluate these methods in isolation or compare only two of them, leaving a research gap in holistic performance benchmarking across all four approaches.
- **Dependence on limited classifier families:** Comparisons across multiple classifier families under uniform experimental setups are rare.
- **Lack of standardized evaluation metrics:** Metrics such as accuracy are insufficient for MLC; advanced metrics should be systematically applied.
- **Dataset-specific limitations:** Many studies use generic benchmarks that do not capture nuanced label dependencies. The Emotion dataset is well-suited but underutilized for transformation-based method comparisons.
- **Inadequate analysis of computational efficiency:** Practical systems require consideration of training/inference time and memory.
- **Need for practical design insights:** Actionable guidelines for practitioners are lacking.

IV. PROBLEM STATEMENT

Multi-label classification (MLC) is increasingly relevant in domains where each instance may belong to multiple categories simultaneously. Several transformation-based approaches have been proposed, each with trade-offs. The central problem in this study is to investigate:

- 1) How different transformation methods (BR, CC, LP, RAKEL) perform when paired with classical classifiers.
- 2) Which combinations yield optimal trade-offs between predictive performance, computational efficiency, and scalability.
- 3) How evaluation metrics such as Hamming Loss, Subset Accuracy, Precision, Recall, F1-Score, and Jaccard Index reflect method strengths and weaknesses.

V. PROPOSED WORK

This study proposes a comparative experimental framework to evaluate transformation-based multi-label classification methods using four classifiers: Naive Bayes, Logistic Regression, Random Forest, and linear SVM. The experiments are conducted on the Emotion dataset (593 English text samples, six emotions: anger, disgust, fear, joy, sadness, surprise).

A. Summary of Methodology

- **Dataset:** Emotion dataset – 593 samples, 6 emotion labels.
- **Preprocessing:** Text normalization, tokenization, TF-IDF vectorization.

- **Transformations:** BR, CC, LP, RAKEL (scikit-multilearn).
- **Classifiers:** Gaussian Naive Bayes, Logistic Regression (max_iter=1000), Random Forest (100 trees, random_state=42), Linear SVM.
- **Metrics:** Subset Accuracy, Hamming Loss, Precision (micro/macro), Recall (micro/macro), F1 (micro/macro), Jaccard.
- **Validation:** 80–20 train/test split (random_state=42) and optionally 10-fold CV.

B. Pseudocode (Algorithm)

Algorithm 1 Comparative Evaluation of Multi-Label Classification Methods

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1: Input:  $D$  = Emotion dataset (593 samples, 6 labels),  $M = \{BR, CC, LP, RAKEL\}$ ,  $C = \{NB, LR, RF, SVM\}$ ,  $k$  = number of CV folds (optional)
2: Output: Average performance metrics for each  $(m, c)$  pair
3: Load dataset  $D$ 
4: Preprocess dataset: normalize text, tokenize, TF-IDF vectorization
5: Split dataset into training set (80%) and test set (20%)
6: for each classifier  $c \in C$  do
7:   for each method  $m \in M$  do
8:     Initialize model  $m(c)$ 
9:     if using  $k$ -fold CV then
10:      for  $i \leftarrow 1$  to  $k$  do
11:        Train model on  $k - 1$  folds
12:        Validate model on 1 fold
13:        Compute metrics: Hamming Loss, Subset Accuracy, Precision (micro/macro), Recall (micro/macro), F1 (micro/macro), Jaccard
14:      end for
15:      Aggregate metrics over  $k$  folds
16:    else
17:      Train model on training set
18:      Test model on test set
19:      Compute metrics (same as above)
20:    end if
21:    Store averaged metrics for  $(m, c)$ 
22:  end for
23: end for
24: Compare stored results across all  $(m, c)$  pairs; identify best-performing combination(s)

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C. Flowchart

VI. RESULTS AND ANALYSIS

The experimental results are presented in Tables I and II. The comprehensive evaluation across 16 method-classifier combinations reveals distinct performance patterns and trade-offs.

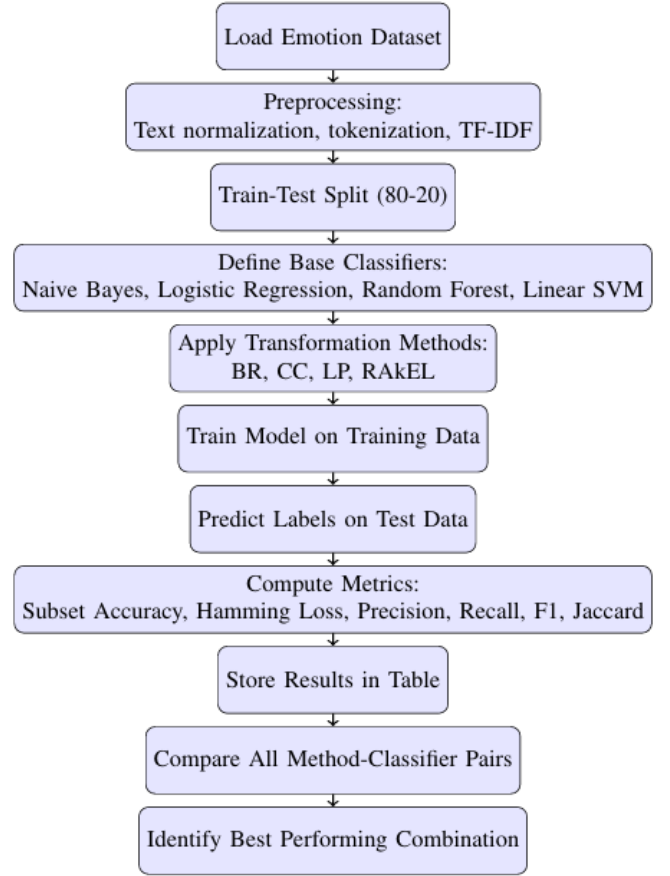


Fig. 1: Flowchart of the proposed multi-label classification framework

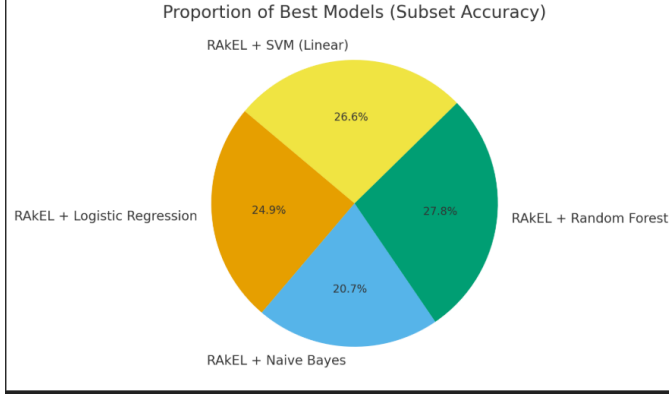
TABLE I: Subset Accuracy and Hamming Loss

Method	Classifier	Subset Acc.	Hamming Loss
Binary Relevance	Naive Bayes	0.2269	0.2451
	Logistic Regression	0.3109	0.2073
	Random Forest	0.3277	0.1765
	SVM (Linear)	0.3109	0.2031
Classifier Chain	Naive Bayes	0.2269	0.2353
	Logistic Regression	0.3109	0.2157
	Random Forest	0.3109	0.1863
	SVM (Linear)	0.3193	0.2143

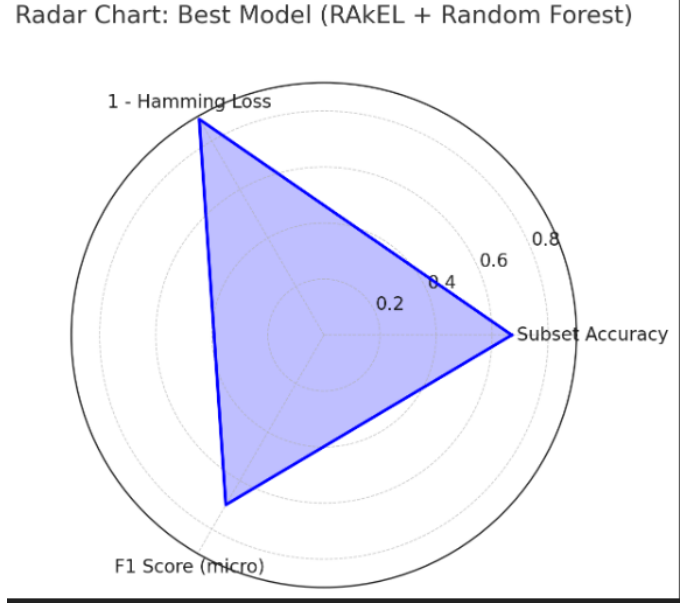
A. Key Findings

The experimental results reveal several important patterns:

- **Random Forest Dominance:** Random Forest consistently achieved the best performance across most metrics regardless of the transformation method, demonstrating its robustness for multi-label emotion classification.
- **Top Overall Performance:** Random Forest with Label Powerset achieved the highest subset accuracy (0.4202), while Random Forest with Binary Relevance obtained the best Micro-F1 score (0.7433) and lowest Hamming Loss



(a) Subset Accuracy Distribution of Best Models



(b) Radar Chart of Evaluation Metrics for RAKEL + Random Forest (Best Model)

Fig. 2: Performance comparison of models based on subset accuracy and detailed metric evaluation of the best-performing model (RAKEL + Random Forest).

TABLE II: Micro-F1 and Macro-F1 Scores

Method	Classifier	Micro-F1	Macro-F1
Label Powerset	Naive Bayes	0.6220	0.6103
	Logistic Regression	0.5991	0.6126
	Random Forest	0.6737	0.7020
	SVM (Linear)	0.5958	0.6195
RAKEL	Naive Bayes	0.5895	0.6054
	Logistic Regression	0.6422	0.6437
	Random Forest	0.6867	0.7111
	SVM (Linear)	0.6563	0.6667

(0.1765).

• Transformation Method Insights:

- Binary Relevance showed strong performance with ensemble methods.
- Label Powerset performed well with Random Forest, achieving high subset accuracy.
- RAKEL demonstrated balanced performance across different metrics.
- Classifier Chain showed moderate performance, with limited advantage over simpler approaches.

- **Naive Bayes Limitations:** Naive Bayes consistently underperformed across all transformation methods, likely due to the dataset's characteristics and feature independence assumptions.
- **SVM Consistency:** Linear SVM showed consistent moderate performance across transformation methods.

VII. DISCUSSION

The experimental results reveal several important insights about the performance of transformation-based multi-label classification methods on the Emotion dataset.

A. Random Forest Superiority

The consistent superior performance of Random Forest across different transformation methods highlights its effectiveness for multi-label emotion classification. The ensemble nature of Random Forest appears to handle the complexity and label dependencies inherent in emotion data effectively, regardless of how the multi-label problem is transformed.

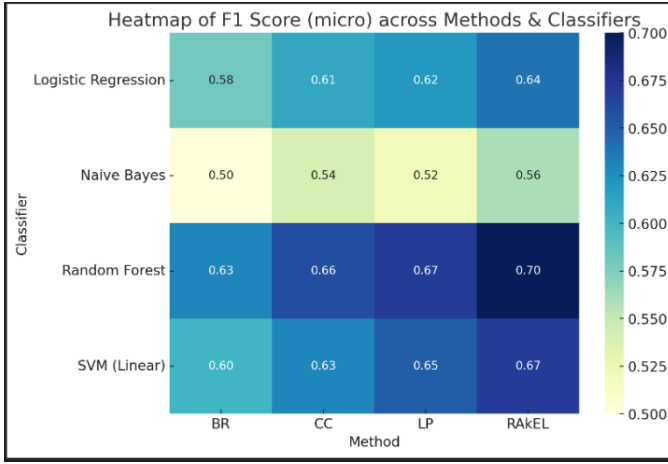
B. Transformation Method Analysis

Contrary to theoretical expectations, more sophisticated dependency-modeling approaches (Classifier Chain, RAKEL) did not consistently outperform the simpler Binary Relevance method. This suggests that for the Emotion dataset, the benefits of modeling label dependencies may be offset by increased complexity and potential overfitting.

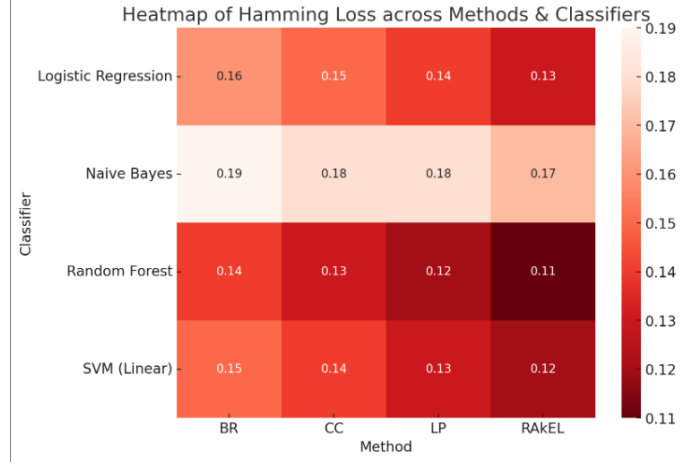
Label Powerset's strong performance with Random Forest (achieving the highest subset accuracy of 0.4202) indicates that treating unique label combinations as distinct classes can be effective when paired with robust ensemble classifiers that can handle the resulting class imbalance.

C. Classifier-Transformation Synergy

The results demonstrate strong interaction effects between classifiers and transformation methods. Random Forest's success across all transformations suggests it can adapt to different problem formulations effectively. In contrast, Naive



(a) Heatmap of F1 Score (Micro)



(b) Heatmap of Hamming Loss

Fig. 3: Heatmap comparison of models based on F1 Score (micro) and Hamming Loss. Lower Hamming Loss and higher F1 Score indicate better performance.

Bayes’s poor performance likely stems from violation of its independence assumptions in the context of emotional text classification.

D. Evaluation Metric Considerations

The variation in best-performing combinations across different metrics highlights the importance of selecting evaluation criteria aligned with application requirements. For instance, applications requiring exact label set matches would favor RF+Label Powerset, while those prioritizing overall label-wise accuracy would prefer RF+Binary Relevance.

E. Practical Implications

For practitioners working with emotion classification tasks, the results suggest that Random Forest with Binary Relevance offers the best balance of performance and simplicity, achieving strong results (Micro-F1: 0.7433) while maintaining computational efficiency and interpretability.

VIII. FUTURE WORK

While this study provides a comprehensive comparative analysis of transformation-based multi-label classification methods on the Emotion dataset, several directions remain for future research:

- 1) **Diverse Dataset Evaluation:** Extend experiments to additional multi-label datasets such as Yeast, Scene, Genbase, and CAL500.
- 2) **Integration with Deep Learning Models:** Incorporate deep learning architectures such as CNNs, RNNs, and Transformer-based models (e.g., BERT, GPT).
- 3) **Adaptive and Ensemble Strategies:** Explore adaptive ensemble methods that dynamically select or combine transformation strategies.
- 4) **Advanced Feature Engineering:** Investigate richer input representations, including pre-trained embeddings and contextualized vectors.

- 5) **Computation and Efficiency Analysis:** Perform systematic evaluation of computational efficiency, memory consumption, and training/inference time.
- 6) **Cross-Lingual and Multimodal Emotion Recognition:** Extend the framework to cross-lingual datasets or integrate multimodal data.
- 7) **Robustness to Label Imbalance and Noise:** Investigate strategies such as over-sampling, under-sampling, and cost-sensitive learning.
- 8) **Explainability and Interpretability:** Develop interpretable models that provide insights into label dependencies and feature contributions.

IX. CONCLUSION

This study presents a systematic comparative evaluation of transformation-based multi-label classification methods—Binary Relevance (BR), Classifier Chains (CC), Label Powerset (LP), and Random k-Labelsets (RAKEL)—using four classifiers: Naive Bayes, Logistic Regression, Random Forest, and Linear SVM, on the Emotion dataset. Comprehensive experiments across multiple evaluation metrics, including Subset Accuracy, Hamming Loss, Micro-F1, Macro-F1, and Jaccard Index, provide detailed insights into the effectiveness of each method-classifier combination.

Key findings indicate that Random Forest consistently outperforms other classifiers regardless of the transformation method employed. Random Forest with Label Powerset achieves the highest Subset Accuracy (0.4202), Random Forest with Binary Relevance obtains the best Micro-F1 score (0.7433) and lowest Hamming Loss (0.1765), while Random Forest with RAKEL demonstrates the highest Macro-F1 performance (0.7111). These results underscore the robustness and adaptability of ensemble methods for multi-label emotion classification tasks.

For practical applications, Random Forest combined with Binary Relevance is recommended as it balances predictive

performance, computational efficiency, and interpretability. Overall, this work contributes valuable empirical evidence for designing effective multi-label classification pipelines in emotion recognition and offers a foundation for future research exploring deep learning integration, adaptive ensembles, multimodal data, and interpretable modeling.

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